

3,500 ft High-Altitude Model Rocket — Full Project Plan

Goal: Build a no-certification-required (G-motor class) model rocket that reaches approximately 3,500 ft, with a button-activated launch system and an onboard flight computer logging velocity, position, and altitude via radio telemetry.

Project Constraints & Ground Rules

- **Motor class:** G-class or smaller (max 160 Newton-seconds total impulse, max 125g propellant, max 1,500g total rocket weight, no metal parts) — this keeps the build in the "Class 1 Model Rocket" category under FAA/NFPA rules, so **no NAR/Tripoli certification is required**.
 - **Radio:** 915 MHz LoRa — license-free in the US, no ham radio exam needed.
 - **Budget approach:** frugal, DIY electronics using generic/clone components rather than name-brand commercial flight computers.
 - **Reusability:** the launch controller and flight computer are designed to be standalone/removable so they can be reused on future rocket builds — only the motor and igniter are single-use.
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Step 1: Model the Rocket in OpenRocket

Before cutting or buying any airframe material, build a full digital model.

- Pick a specific motor: a 29mm G motor near the top of the impulse range (100–150 Ns). Look up exact thrust curve and total impulse data on **ThrustCurve.org** — this data can be imported directly into OpenRocket.
- Input your **actual electronics weight** (flight computer + GPS + radio + battery) into the model — this is the single biggest factor separating a real design from an optimistic guess.
- Iterate on body tube diameter, fin size, and mass until you get:
 - A simulated apogee around 3,500 ft
 - A stability margin of 1–2 calibers (center of gravity ahead of center of pressure by 1–2 body-tube diameters)
- OpenRocket will calculate CG/CP automatically as you adjust the design — use this to converge on a safe, stable shape.

Why this step is first: every later decision (tube diameter, how much room electronics have, how much the recovery system needs to handle) flows from this model. Skipping simulation means guessing, and guessing at this altitude means either an unstable rocket or a big miss on your altitude target.

Step 2: Order Materials and Electronics

Order in this sequence to manage both budget and lead time:

1. **Electronics first** (longest shipping lead time):
 - Microcontroller (Arduino Nano clone or similar)
 - Barometric pressure sensor (BMP280/BMP388) — for altitude
 - Accelerometer/gyro (MPU6050) — for velocity (via sensor fusion with the barometer)
 - GPS module (NEO-6M or similar) — for position, and a second altitude/velocity source
 - MicroSD card module — for onboard data logging (backup if radio has gaps)
 - Small LiPo battery + voltage regulator
 - Two RFM95 LoRa modules (one for the rocket, one for the ground station)
 - Source: **AliExpress** (cheapest, 2–4 week shipping) or **Amazon** (pricier, faster)
2. **Airframe materials second** (once your OpenRocket model is finalized):
 - Phenolic paper body tube (stronger than basic cardstock, still lightweight — avoid fiberglass, which is overkill weight for G-motor power levels)
 - Fin stock: basswood or thin (1/8") plywood (tougher than balsa, only a small weight penalty)
 - Plastic nose cone (durable, lightweight, and gives GPS a clean position near the top of the airframe)
 - Source: **Apogee Rockets** or **Wildman Rocketry**
3. **Insurance spares** — order 1–2 extra of the cheapest failure-prone parts: a spare microcontroller (easy to fry with a wiring mistake), a couple of extra igniters, extra jumper wires/connectors. This is ~\$10–15 extra but avoids a shipping delay killing momentum right before a launch attempt.
4. **Motor last**, close to your actual launch date — no reason to store propellant for weeks.

Vendor Reference

Vendor	Use for
Apogee Rockets	Body tubes, nose cones, fin stock, motors, recovery gear
Wildman Rocketry	Price comparison for airframe parts and motors

Estes Rockets	Basic tubes/nose cones/low-power motors if prototyping first
ThrustCurve.org	Motor thrust curve and impulse data for OpenRocket
Amazon	Fast electronics shipping
AliExpress	Cheapest electronics, slower shipping

Step 3: Build and Bench-Test the Flight Computer

This is the core of the telemetry system — velocity, position, and altitude all originate here.

- Wire the **barometer** to the microcontroller alone first. Confirm the altitude reading changes when you change floors or blow gently across the sensor.
- Wire the **accelerometer/gyro** alone. Confirm it responds when tilted or shaken.
- Wire the **GPS module** alone. Confirm it can get a satellite fix (usually needs to be near a window or outdoors).
- Wire the **SD card module** alone. Confirm it can write and read a test file.
- Once each works independently, connect **all four together** and confirm the microcontroller can read all sensors and log to SD simultaneously without conflicts (this is where most real bugs surface — shared pins, power draw issues, timing conflicts).

Why test in isolation first: a dead or miswired module costs \$5 and 10 minutes to diagnose on a breadboard. The same problem discovered after the electronics are epoxied into a rocket costs you the whole flight.

Step 4: Build and Bench-Test the Radio Link

- Set up the two RFM95 LoRa modules — one destined for the rocket, one for the ground station.
 - Confirm you can send a message from one to the other across a real-world distance (your yard, your neighborhood) *before* it's anywhere near the airframe.
 - This validates range and reliability while it's still easy to debug — sitting still at a workbench, not troubleshooting mid-flight.
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Step 5: Build and Bench-Test the Launch Controller

Build this as a **standalone box**, physically separate from the rocket, so it can be reused on every future launch.

- **Arm/safety key switch** — keeps the circuit electrically dead until you're ready to launch.
- **Momentary launch button** — completes the circuit to fire the igniter.
- **Continuity-check LED** — lights up only when the igniter circuit is complete, so you know it will actually fire before you press the button.
- **12V battery** (or similar) — supplies firing current.
- **Long lead wires with alligator clips** (~20–30 ft) — connects the controller to the igniter on the pad, letting you stand a safe distance away.

Test this against a dummy resistor (standing in for a real igniter) until you're confident:

- The LED only lights when armed and properly connected
- The circuit only fires when the key is turned AND the button is pressed

Reusability note: because this uses simple clip-lead connections, it works with *any* future rocket's igniter — you build this circuit exactly once.

Step 6: Write the Software and Run a Full Integrated Dry Test

- Write firmware that:
 - Reads all sensors (barometer, accelerometer, GPS)
 - Fuses barometer + accelerometer data into a more accurate combined velocity/altitude estimate than either sensor alone
 - Formats and transmits this data over the LoRa radio
 - Logs everything to the SD card as a backup
- Write matching ground-station code/display to receive and show the incoming telemetry in real time.
- Run one **full dry run**: power on the complete assembled system, simulate a "flight" by walking around with it, and confirm:
 - Sensors are all logging correctly
 - Radio telemetry is being received on the ground station
 - SD card backup log matches what was transmitted

This is your last full-system check before anything gets permanently mounted in the airframe.

Step 7: Assemble the Rocket and Ground-Test Recovery

- Build the physical airframe: motor mount, fins, nose cone.
 - Build the recovery system: parachute or streamer + shock cord, sized for a safe descent rate (roughly 15–20 ft/s landing speed).
 - Mount the flight computer, GPS, radio, and battery on a **removable sled or tray** (not permanently soldered into the airframe) — this lets you pull the whole electronics package out and reuse it in a future rocket build.
 - **Ground-test the recovery deployment:** with the real igniter installed, confirm continuity, then test-fire the ejection charge to confirm the parachute/streamer actually deploys cleanly — don't find out about a packing problem on the real flight.
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Step 8: Launch Day — Prep, Flight, Recovery, and Review

1. **Site selection:** an open field with generous clearance, no overhead obstructions, and no nearby air traffic.
 2. **Pre-flight checklist:**
 - Continuity check confirms the igniter circuit is live
 - Arm switch tested
 - Radio link confirmed working in both directions
 - Recovery system packed correctly
 3. **Launch:** insert the safety key, countdown, press the button.
 4. **Track:** follow the descent using the live radio telemetry (position/altitude/velocity).
 5. **Recover:** walk to the GPS-reported landing coordinates to retrieve the rocket.
 6. **Review:** pull the logged data off the SD card and compare it against your original OpenRocket prediction — this tells you how close your simulation was to reality and what to adjust for the next build.
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Quick Reference: Estimated Budget

Category	Estimated Cost
Flight computer electronics (microcontroller, sensors, GPS, SD, battery)	\$25–45
Radio (2x LoRa modules + ground receiver)	\$15–30
Launch controller (switches, battery, wiring)	\$15–30
Airframe materials (tube, fins, nose cone)	\$20–40

Motor + igniter	\$15–25
Insurance spares	\$10–15
Total (first build)	~\$100–185
Cost per future launch (reusing controller/flight computer)	~\$15–25

Generated as a project plan based on a conversation about designing a no-certification-required, button-launched, telemetry-equipped model rocket targeting 3,500 ft altitude.